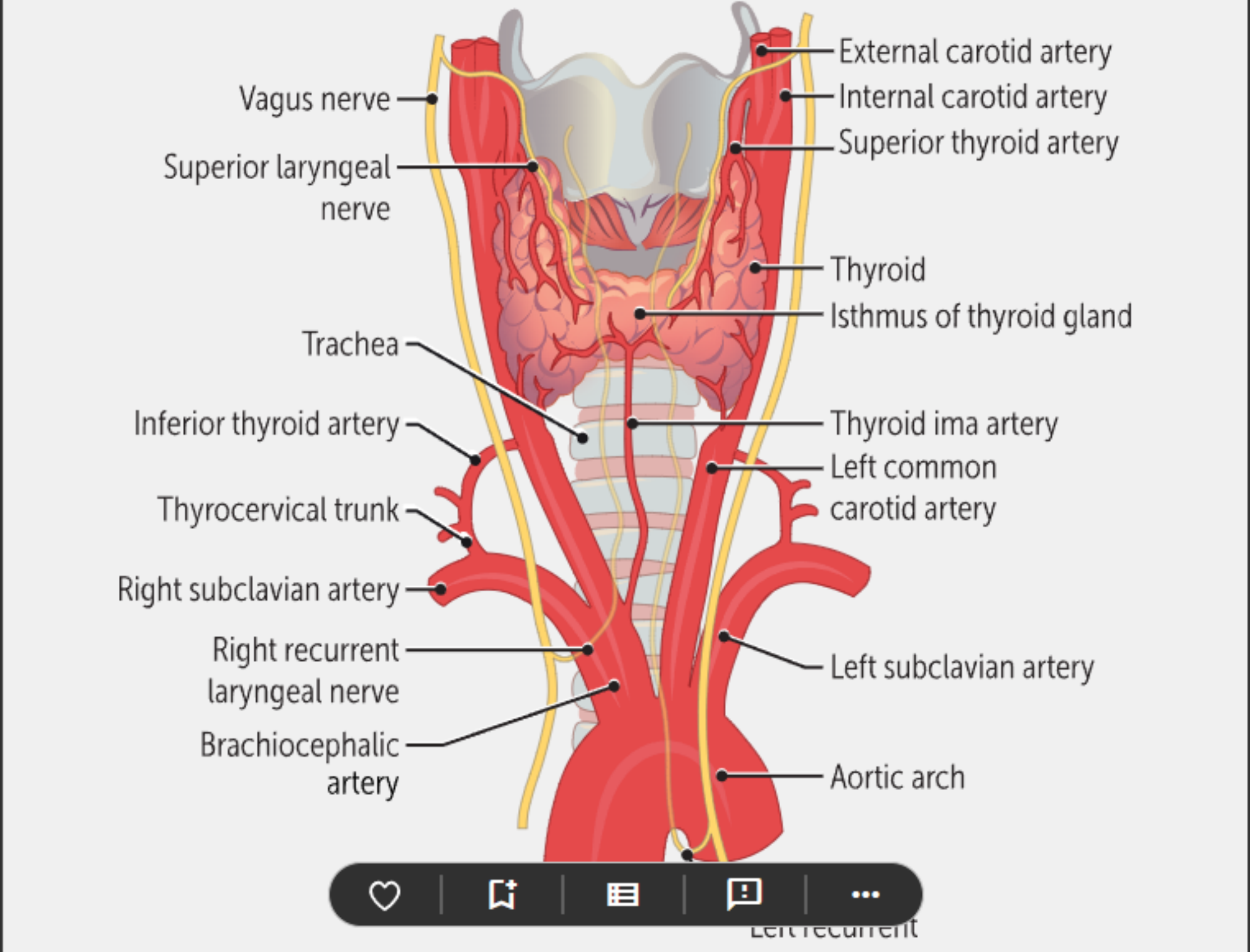


arteriosus.

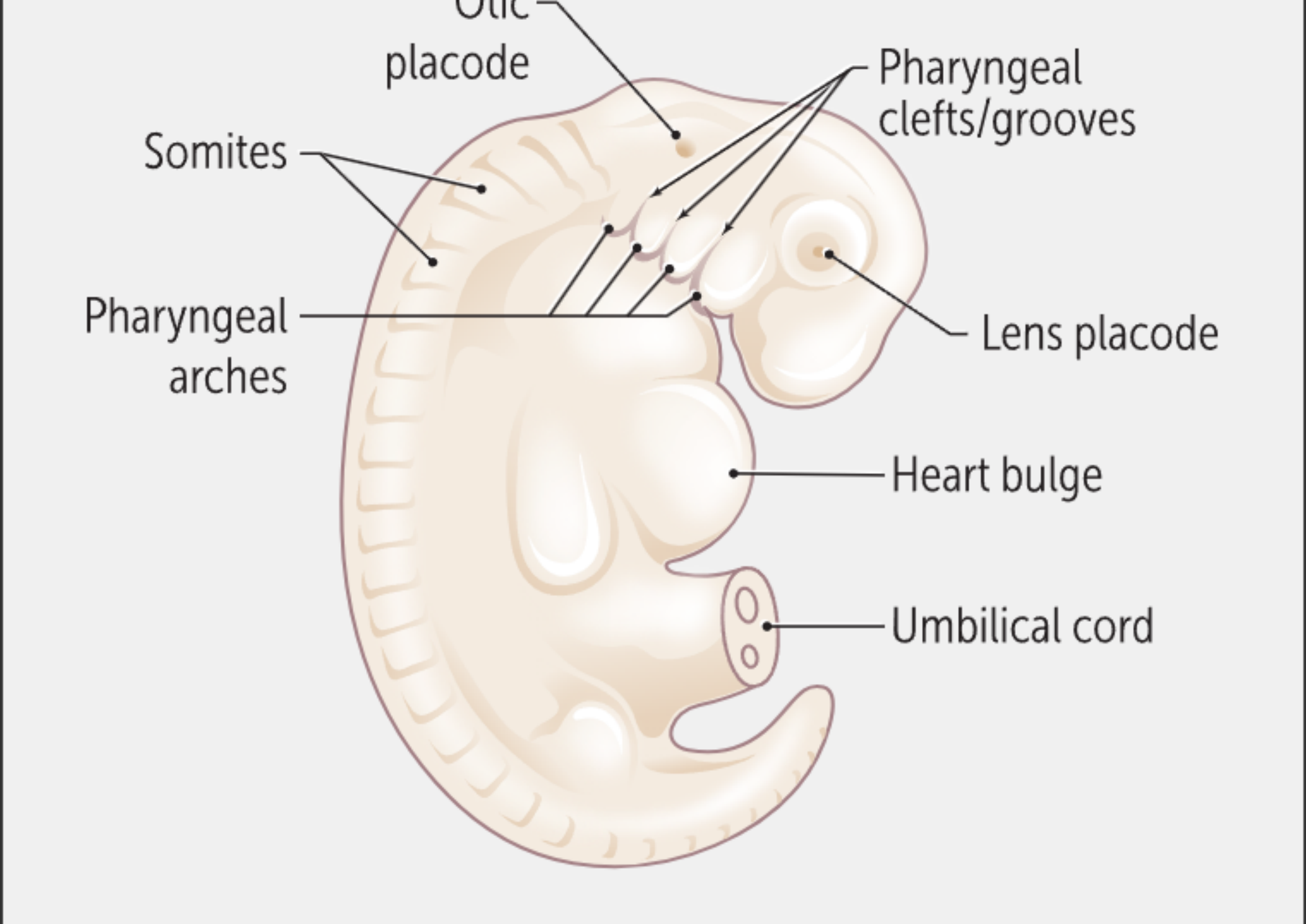
Adult Aortic Anatomy

As shown in [Figure 1](#), the adult aorta gives rise to several major branches whose asymmetric arrangement reflects their embryologic origin. On the **right**, a brachiocephalic trunk divides into the right subclavian and right common carotid arteries. On the **left**, the common carotid and subclavian arteries arise separately from the aortic arch.



- **Artery** (called an aortic arch)

These embryonic aortic arches should not be confused with the single adult aortic arch, which is only one of their many derivatives.



QUIZ

Figure 3

Six pairs of aortic arches form sequentially from head to tail, although the **fifth pair** soon regresses. These initially **symmetric vessels** remodel extensively, establishing the framework for the **asymmetric great arteries** of the head, neck, and thorax seen in the adult.

Major arteries of the head and neck arise from which embryonic arteries?

How Does Each Aortic Arch Develop into the Adult Pattern? (LO#1)

As development proceeds, the aortic arches undergo selective **regression, persistence, and remodeling** to produce the

QUIZ

Figure 4

What does the third aortic arch form?

Fourth Aortic Arches. The right side becomes the proximal right subclavian artery, while the left side forms the arch of the aorta between the left common carotid (LCCA) and left subclavian arteries (LSA). [Figure 4 & 5](#).

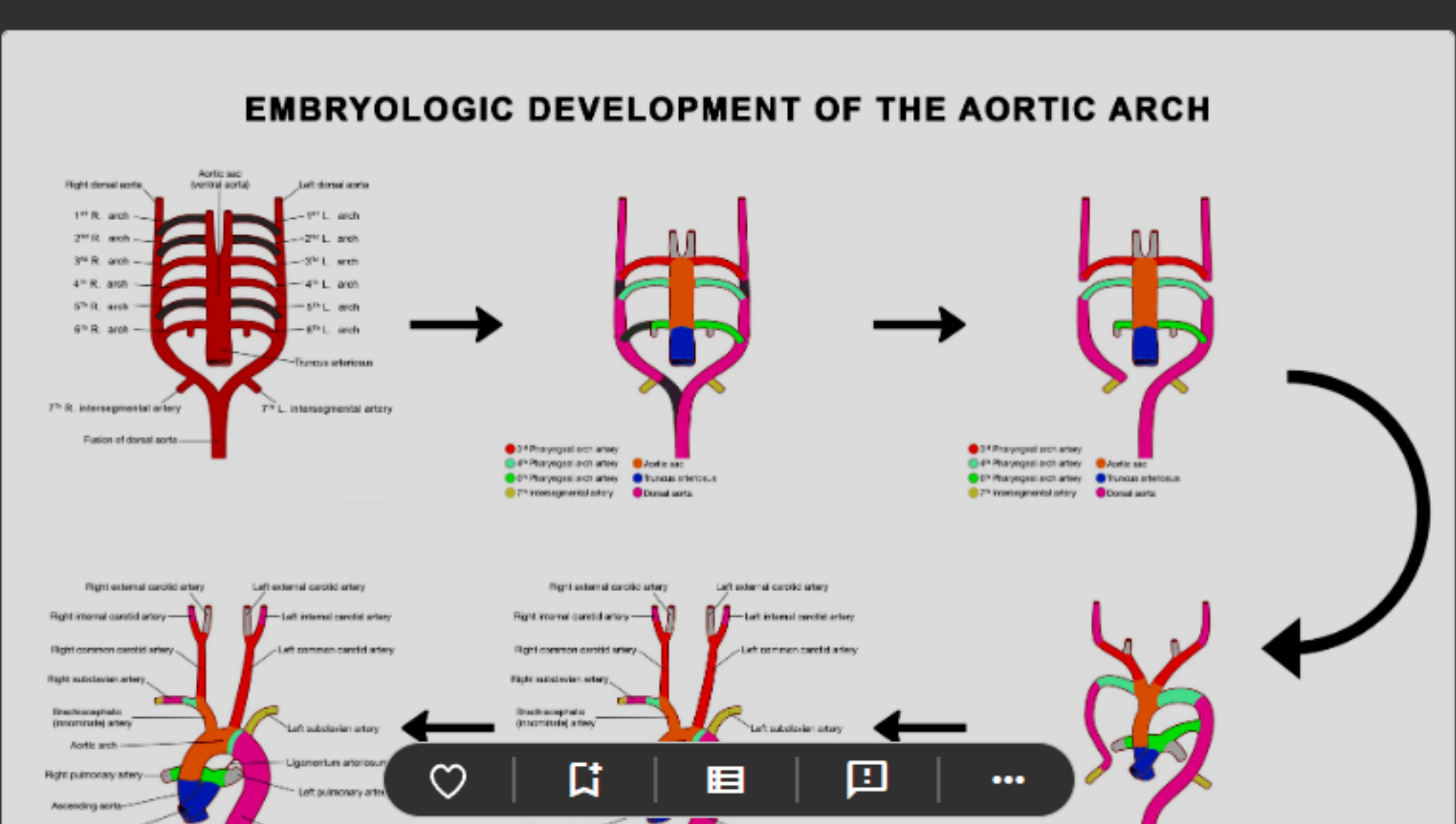
Sixth Aortic Arches | right and left pulmonary arteries. On the left, the distal segment persists as the **ductus arteriosus** (LSA)

5th	Does not form	Does not form
6th	Proximal right pulmonary artery	Proximal left pulmonary artery and ductus arteriosus

Dorsal Aorta Derivatives

As the aortic arches remodel, the paired dorsal aortae also undergo major changes ([Figure 5](#)).

- Caudally, the two dorsal aorta fuse to form a single descending aorta.
- Cranially, the right dorsal aorta partially regresses, leaving the distal segment as part of the right subclavian artery, while the left dorsal aorta remains as the continuation of the descending aorta.



We can't discuss development of the aortic arches and their derivatives without also considering the nearby recurrent laryngeal nerves. Because these nerves develop in close proximity to the aortic arches, their final positions are shaped by the same vascular remodeling events. Let's examine how these nearby processes influence one another.

How Do the Recurrent Laryngeal Nerves Develop? (LO#2-3)

The **recurrent laryngeal nerves** are branches of the **vagus nerve** (cranial nerve X) that supply all the **intrinsic muscles** of the **larynx**, except the **cricothyroid**. In the adult, their **right and left courses** differ markedly because of changes in the surrounding **aortic arches** during development.

- The **right recurrent laryngeal nerve** loops under the right subclavian artery before ascending toward the larynx.
- The **left recurrent laryngeal nerve** loops under the arch of the aorta, just posterior to the ligamentum arteriosum.

This asymmetry reflects differences in the **sixth aortic arch remodeling**, which will be explored next.

Recurrent Laryngeal Nerve Asymmetry

During early development, the **right and left recurrent laryngeal nerves** both loop around the **sixth aortic arch** ([Figure 6](#)).

As the embryo elongates, the right recurrent laryngeal nerve moves upward and ultimately loops around the **right subclavian artery**, a derivative of the fourth aortic arch.

- On the **left**, the **sixth aortic arch persists** as the **ductus arteriosus**, maintaining its connection to the aortic arch. The left recurrent laryngeal nerve therefore remains looped under the **aortic arch**, adjacent to the **ligament arteriosum**.

Why do the right and left recurrent laryngeal nerves follow asymmetric courses?

CLINICAL CORRELATION
Because of their close relationship to the **thyroid and parathyroid glands**, the **recurrent laryngeal nerves** are vulnerable during **neck surgeries**, especially thyroidectomy and parathyroidectomy.

- The embryonic aortic arches are paired arteries within the pharyngeal arches that remodel to form the major arteries of the head, neck, and thorax.
- Although six pairs initially appear, the **fifth pair regresses**, and selective persistence of others produces the asymmetric pattern of the adult great vessels.
- The **left sixth aortic arch** forms the **ductus arteriosus**, while the corresponding right arch regresses - an event that explains the different paths of the **right and left recurrent laryngeal nerves**.
- Understanding these remodeling processes clarifies both normal anatomy and clinical conditions such as **coarctation of the aorta** and **recurrent laryngeal nerve injury**.

Review Questions

1. Which aortic arch is responsible for the formation of the ductus arteriosus?

- A. ☐ First
- B. ☐ Second
- C. ☐ Third
- D. ☐ Fifth
- E. ☒ Sixth

3. Which developmental event causes the **right recurrent laryngeal nerve** to loop under the **right subclavian artery** rather than the aortic arch?

- A. ☒ Degeneration of right sixth arch
- B. ☐ Asymmetry in aortic arch development
- C. ☐ Regression of the left sixth aortic arch
- D. ☐ Fusion of the dorsal aortae

Hide Explanation

The correct answer is degeneration of the right sixth aortic arch (A). This degeneration pulls the right recurrent laryngeal nerve superiorly. Asymmetry in sixth aortic arch development (B) affects the course but does not directly cause the looping. Regression of the left sixth aortic arch (C) does not occur. Fusion of the dorsal aortae (D) occurs below the region where the recurrent laryngeal nerves loop and does not affect the relative position of the nerves.

How would you rate this Brick?